

STM32G431 Motor Controller: Master Technical Handover

Project Phase: Diagnostic / Sensorless ESC Transition **Engineer:** CK

1. Physical Layer & Component Engineering

Isolation & Gate Drive Logic

- **Gate Drive Strategy:** Driving the IR2104 by connecting **IN** and **SD** pins together to a single PWM signal.
 - *Benefit:* Eliminates separate dead-time logic from the MCU; the IR2104 provides its own internal dead-time (~520ns).
- **Optocouplers (4 Total):**
 - **3x 6N137:** High-speed digital optos for Phase signals. **Note: These are inverting.** STM32 `LOW` = Opto LED ON = Opto Output LOW = IR2104 `IN/SD` HIGH.
 - **1x PC817:** Global Shutdown/Enable path (Active High).
- **Gate Drivers:** 3x IR2104.
 - *The XNOR Discovery:* During `test8`, we found previous chip batches acted like XNOR gates (requiring specific signal mismatches to enable). We are transitioning to new hardware to restore standard behavior.
- **Bootstrap Circuit:** Standard diode + capacitor per phase. Requires periodic low-side switching (LO ON) to maintain high-side gate charge.

Signal Conditioning & Filtering

- **RC Filters:** 15.6kHz Low-Pass filter on every internal OPAMP non-inverting input (PBO, PA1, PA7).
- **Voltage Clamping:** Schottky diodes connected to 3.3V rails on each OPAMP input pin to protect the STM32 from high-voltage back-EMF spikes.
- **Decoupling:** 0.1uF and 1uF ceramic capacitors distributed across all IC supply pins (MCU, Drivers, OPAMPs).
- **Snubbers:** RC snubbers placed across MOSFET Drain-Source to suppress high-frequency switching ringing.

2. Verified Code Snippets (The "Hard-Learned" Blocks)

Working CORDIC Interface (Fixed)

Crucial bit-masking for the `NRES` (2 results) and `PRECISION` (6 iterations) flags.

```
inline void get_Fast_Sin_Cos(float angle_rad, float* out_sin, float* out_cos) {  
    while(angle_rad > PI) angle_rad -= 2.0f * PI;
```

```

while(angle_rad < -PI) angle_rad += 2.0f * PI;
int32_t angle_q31 = (int32_t)((angle_rad / PI) * 2147483648.0f);
CORDIC->WDATA = angle_q31;
// First read is Cosine, second is Sine
*out_cos = (float)((int32_t)CORDIC->RDATA) / 2147483648.0f;
*out_sin = (float)((int32_t)CORDIC->RDATA) / 2147483648.0f;
}
// Init in setup():
// CORDIC->CSR = (6 << CORDIC_CSR_PRECISION_Pos) | CORDIC_CSR_NRES;

```

Dynamic Calibration of OPAMP Reference Points

Automatically identifies the zero-current bias point of the internal OPAMPs in real-time.

```

static float maxU = -10.0, minU = 10.0;
float decay = 0.002f;
if (ampsU > maxU) maxU = ampsU; else maxU -= decay;
if (ampsU < minU) minU = ampsU; else minU += decay;
float centerU = (maxU + minU) / 2.0f;
float finalU = ampsU - centerU; // The pure AC component for FOC/ZCD

```

SVPWM / Vector Phase Output

Generating 3-phase voltages from Clarke/Park variables (V_α, V_β).

```

int offset = 4250; // ARR/2 for 170MHz @ 20kHz
TIM1->CCR1 = offset + (int)(V_alpha);
TIM1->CCR2 = offset + (int)(-0.5 * V_alpha + 0.866 * V_beta);
TIM1->CCR3 = offset + (int)(-0.5 * V_alpha - 0.866 * V_beta);

```

3. Hardware Mapping (STM32G431CBU6)

- **PWM (TIM1):** PA8 (U), PA9 (V), PA10 (W).
- **Current Sensing (PGA Gain 16x):**
 - **Phase U:** PB0 (OPAMP3)
 - **Phase V:** PA1 (OPAMP1)
 - **Phase W:** PA7 (OPAMP2)
- **Global Shutdown:** PB9 (SD).

4. Critical Operational Lessons

1. **Blanking Window:** Inductive flyback spikes must be ignored for the first ~15% of a step to avoid "fake" Zero Crossings in sensorless mode.
2. **ADC Power State:** The STM32G4 requires a specific sequence: ADVREGEN -> 20us Wait -> ADEN -> ADRDY check.

3. **V/f Relationship:** PWM duty must scale linearly with frequency to maintain torque against increasing back-EMF at high RPM.